Physics - Graduate Level Assessment 18 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the jovian planets have rings?
   (a) Neptune
   (b) Uranus
   (c) Saturn
   (d) all of the above

2. All of the following have been cited as evidence for the Giant Impact Hypothesis except:
   (a) The Moon and Earth have nearly identical isotopic compositions.
   (b) The Moon's orbit is not in the same plane as Earth's equator.
   (c) The Moon's crust is thicker on the far side than the near side.
   (d) The Moon has a much smaller iron core than the Earth even considering its size.
   (e) The Moon was once entirely molten.

3. Across what potential difference in V does an electron have to be accelerated to reach the speed v = 1.8 x 10^7 m/s? Calculate this relativistically.
   (a) 750.0
   (b) 924.0
   (c) 1300.0
   (d) 800.0
   (e) 870.0

4. When a wave moves from shallow water to deep water, the
   (a) frequency does not change, the wavelength increases, and the speed increases
   (b) frequency decreases, the wavelength increases, and the speed does not change
   (c) frequency increases, the wavelength increases, and the speed decreases
   (d) frequency increases, the wavelength does not change, and the speed increases
   (e) frequency increases, the wavelength decreases, and the speed does not change

5. Find the energy equivalent to 1 amu.
   (a) 1000 Mev
   (b) 650 MeV
   (c) 800 MeV
   (d) 931 Mev
   (e) 500 Mev

True / False (10 marks)

Answer True or False

6. True or False: The energy of a photon with a wavelength of 400 nm is approximately 3.1 eV.
7. True or False: The value of g in Paris can be approximated as 33.000 ft/s^2 based on the pendulum's observed behavior.
8. True or False: The solar cycle, characterized by an increase in sunspot number, typically occurs every 11 years.
9. True or False: Plating a spoon at 0.50 ampere for 20 minutes results in a deposition of 1.000 grams of silver.
10. True or False: The time required for a satellite to complete one revolution is directly proportional to the mass of the planet it orbits.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the significance of the U.S.S.R.'s lander missions to Venus in the context of planetary exploration and their contributions to our understanding of the planet's atmosphere and geology.
12. A box with a mass of 10 kg slides over a rough surface at a constant velocity of 2 m/s under the influence of a 5 N horizontal force. Calculate the work done by this force over 1 minute, and discuss the implications of friction in this scenario.

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